

Meshtastic Example in Emergency Communication

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Abstract

The potential damage, loss, and injury from natural events, timing and location of which are uncertain, can lead to unpredictable natural disasters, posing significant threats to humanity. As a result of these disasters, people lose their lives or are injured, and living spaces become unusable or inoperable for extended periods. Immediately after these natural disasters, it is very important for aid teams to maintain uninterrupted communication with each other and with crisis management centers established in the disaster area in order to deliver emergency medical and humanitarian aid to injured and in need. Existing communication systems can become unusable, either partially or completely due to the impact of a natural disaster, or even if undamaged, due to uncontrolled post-disaster communication traffic. These adverse conditions cause significant difficulties in communicating with emergency medical aid, search, and rescue teams. For this purpose, this study utilizes using LoRa radio transceivers communication nodes developed by a project called Meshtastic, which utilizes radio frequencies and does not utilize existing communication infrastructure. LoRa radio transceivers use the ISM band, which is used for Industrial, Scientific and Medical Devices and does not require any licensing in many countries, including ours. These handheld nodes can conduct encrypted or unencrypted written communication among themselves. Furthermore, the coordinates of the nodes can be tracked using GPS receivers, and surrounding nodes can be tracked as well. Furthermore, the modules' built-in Wi-Fi hardware connects to the system's internet infrastructure, enabling all handheld nodes in the network to communicate simultaneously, either individually or in their entirety. All terminals in the network can be tracked on a map via the Meshtastic application and communicated through the application. In this study, three nodes operating at 433 MHz and 868 MHz were created, enabling communication with each other from different locations via both radio frequency and the Meshtastic application. This study successfully tested Meshtastic, an alternative communication method that can be used in extraordinary situations such as natural disasters by communicating with Meshtastic nodes without relying on the existing communication infrastructure that is used in daily life. The main objective of this study was to successfully establish communication between nodes using the ISM band, without requiring any internet infrastructure or specific usage permits or licenses.

Keywords: “Meshtastic, LoRa, disaster, communication.”

1. Introduction

A natural disaster is a large-scale hazard or event that occurs largely or completely beyond human control and can cause loss of life and property [1]. Humanity is put in jeopardy by natural disasters caused by natural events such as earthquakes, floods, inundations, landslides, rockfalls, avalanches, storms, tornadoes, and volcanoes, as well as by human-induced disasters such as major industrial accidents. The timing and location of these natural and human-caused disasters, the extent of injuries, loss of life, and damage they will cause, are unpredictable. Due to this nature of disasters, it is impossible to intervene when they occur and mitigate their undesirable consequences [1].

After a disaster occurs, it is very important for local authorities to communicate rapidly and continuously with the central authority. Furthermore, it is also necessary for local authorities to communicate with each other and with established crisis management centers, including search, rescue, emergency medical response, humanitarian aid teams, debris removal, and logistics support teams for disaster victims. In emergency communications, existing communications infrastructure can become unusable due to the negative effects of a disaster, either completely or partially damaged, or even if the infrastructure is not damaged, due to uncontrolled communication traffic, rendering the existing communications infrastructure unusable for teams. This study utilizes the Meshtastic communications platform, which does not utilize existing communications infrastructure.

A literature review on Meshtastic revealed two significant studies. In their study, J. Dhanasekar et al. created LoRa nodes to address the explosion risks posed by high-pressure steam, combustion gases, chemicals, and moving machinery, which create a dangerous environment for workers. These LoRa nodes detect the dangerous gas levels inside boilers and inform the

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maintenance team and workers when the levels exceed a predetermined range [2]. In a study carried out by Liang-Bi Chen et al., it was found that in Taiwan, which has one of the highest mountain ranges in the world, mountain-related accidents are becoming increasingly common among mountain hikers. To facilitate mountain rescue operations, physiological sensors providing real-time health monitoring are attached to the arms of mountain hikers. Data obtained from these sensors is transmitted via Bluetooth modules to LoRa modules and then to the Meshtastic application. This allows for easy location tracking of the injured person on a map using GPS receivers in the event of an accident [3].

1.1. What is Meshtastic?

Meshtastic is an open-source project that uses LoRa radio transceivers as a long-range off-grid communications platform without relying on existing communication channels. Meshtastic utilizes LoRa radios, a radio protocol that, unlike amateur radio operations, does not require additional licenses or certifications [4-6]. Figure 1 illustrates the popular Lilygo T-Beam LoRa development board used for the Meshtastic application. This board includes built-in GPS, a LoRa module, and a display [7].

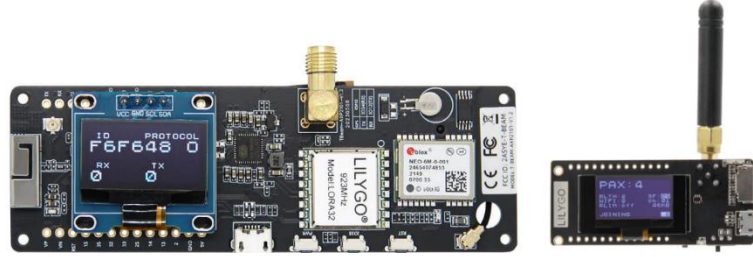


Fig. 1. Various nodes with Meshtastic support

A Meshtastic network refers to a cluster of nodes that share the same LoRa spreading factor, frequency, and bandwidth. A node can be a member of a single radio network and can only receive and send messages within its own network. A maximum of eight channels can be created within a radio, and each channel can send and receive encrypted or unencrypted messages. Meshtastic radios are called nodes. Nodes are designed to form a network by rebroadcasting messages received by them. This method ensures that all members of the network receive the message. Figure 2 shows the communication topology between Meshtastic nodes themselves and between nodes and mobile devices such as tablets, phones, and computers [4, 8, 9].

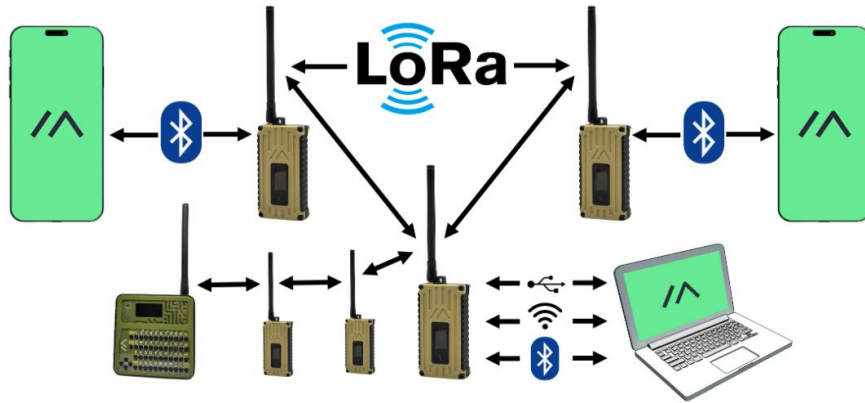


Fig. 2. Meshtastic Network Topology

If desired, a node can connect to the internet via Wi-Fi and receive messages from the network, or connect to MQTT servers developed for this purpose, and send messages to or receive messages from the network. When a message is sent directly to the network from a node, then it is delivered to another node connected to the same channel over the Meshtastic network. When a message is sent to the network through the Meshtastic application, then it is delivered to the radio via Bluetooth, Wi-Fi, Ethernet, or serial port. This message is delivered to the Meshtastic network via the relevant radio. If no other nodes return a confirmation message within the specified timeout, then the message is repeated up to three times. When a node in the network receives a message packet, then it checks whether it has already received the message. If the node has received the message, then it ignores that message; if not, then the node broadcasts the message so that the most distant nodes can receive it. Figure 3 shows the hop propagation in Meshtastic nodes [4, 10].

Thus, a sent message is delivered from the closest node to the farthest node. Each message has a defined hop count. This count is decreased by one as it hops from one node to the next, and the message is stopped when the hop count reaches 0 [4].



Fig. 3. Meshtastic hop propagation

1.2. What is LoRa?

LoRa, named after the English words “Long Range”, is a radio transceiver operating at radio frequencies that uses chirp spread-spectrum communication modulation [11]. LoRa devices were first developed by Cycleo, a French company, and acquired by Semtech in 2012 and patented in 2014 [12]. LoRa radio devices are battery-powered, low-power devices that offer a solution for communicating with each other over long distances [13-17]. Figure 4 shows the AI Thinker Ra-01 module, which is popularly used in LoRa and Meshtastic nodes [18].



Fig. 4. LoRa Radio Modules

1.3. Meshtastic Radio Settings

LoRa nodes use the 433MHz (SX1278) and 868MHz (SX1262) frequencies for the European frequency bands and the 915MHz (SX1276) frequency for the North American frequency band. The maximum output power of the nodes is +10dBm for 433MHz, +27dBm for 868MHz, and +30dBm for 915MHz [19, 20].

There are eight different predefined modes for LoRa radios, as presented in Table 1. For custom configuration, they can be selected through the Meshtastic app. For fast transmission over long distances, the LongFast preset should be selected. This will enable data transmission at a rate of 1.07 kbps over a 250kHz bandwidth [4].

Table 1 presents the bandwidth and data rates for the eight available presets. LoRa radios from many manufacturers support the Meshtastic network. Table 2 shows a comparison of popular LoRa radio modules that support Meshtastic. The cheapest, most accessible, and most widely used of these are the RAK 4631 [21], LILYGO TTGO T-Beam [7], LILYGO® TTGO T-Echo [22], HELTEC® LoRa 32 Wireless Stick Lite V3 [23], HELTEC LoRa32 V3/V3. 1 [24], and Wireless Tracker v1.0 [25].

Table 1. Bandwidth and data rates for 8 presets available for selection.

<i>Radio Preset</i>	<i>Alt Preset Name</i>	<i>Data-Rate</i>	<i>SF / Symbols</i>	<i>Coding Rate</i>	<i>Bandwidth</i>	<i>Link Budget</i>
Short Range / Turbo	Short Turbo	21.88 kbps	7 / 128	4/5	500 kHz	140dB
Short Range / Fast	Short Fast	10.94 kbps	7 / 128	4/5	250 kHz	143dB
Short Range / Slow	Short Slow	6.25 kbps	8 / 256	4/5	250 kHz	145.5dB
Medium Range / Fast	Medium Fast	3.52 kbps	9 / 512	4/5	250 kHz	148dB
Medium Range / Slow	Medium Slow	1.95 kbps	10 / 1024	4/5	250 kHz	150.5dB
Long Range / Fast	Long Fast	1.07 kbps	11 / 2048	4/5	250 kHz	153dB
Long Range / Moderate	Long Moderate	0.34 kbps	11 / 2048	4/8	125 kHz	156dB
Long Range / Slow	Long Slow	0.18 kbps	12 / 2048	4/8	125 kHz	158.5dB

Table 2 shows the microcontrollers used in Meshtastic modules, including LoRa radios, built-in wireless connectivity, Bluetooth, and a Global Positioning System (GPS) [4].

Table 2. Comparison of Meshtastic modules

Name	MCU	Radio	Wi-Fi	BT	GPS
RAK4631	nRF52840	SX1262	NO	5	NO*
LILYGO TTGO T-Beam	ESP32-S3	SX1262	2.4GHz b/g/n	5	YES
LILYGO TTGO T-Echo	nRF52840	SX1262	NO	5	YES
HELTEC LoRa32 T3-S3	ESP32-S3	SX1262 SX1276 SX1280 LR1121	2.4GHz b/g/n	5	NO*
HELTEC LoRa32 V3/3.1	ESP32-S3FN8	SX1262	2.4GHz b/g/n	5	NO*
HELTEC Wireless Stick Lite V3	ESP32-S3FN8	SX1262	2.4GHz b/g/n	5	NO*
HELTEC Wireless Tracker	ESP32-S3FN8	SX1262	2.4GHz b/g/n	5	YES

*Can be added externally.

1.4. Module Settings

The Meshtastic radios can optionally be equipped with LED indicators, buzzer, and vibration motors to notify users of incoming messages, an OLED display to display incoming messages, a microphone to send voice messages, a speaker to listen to incoming messages, a button to send predefined text messages, and a keyboard for direct message entry. The functionality of the radio can be enhanced by configuring the Meshtastic application and integrating it into the node [4].

2. Materials and Methods

In this study, one LILYGO TTGO T-Beam and two HELTEC Wireless Stick Lite V3 modules were used for two different Meshtastic networks operating at 433MHz and 868MHz. This created three nodes operating at 433MHz and 868MHz. Since the LILYGO TTGO T-Beam nodes have a built-in OLED display and GPS receiver, only the M5STACK CardKB [26] keyboard is included in these nodes.

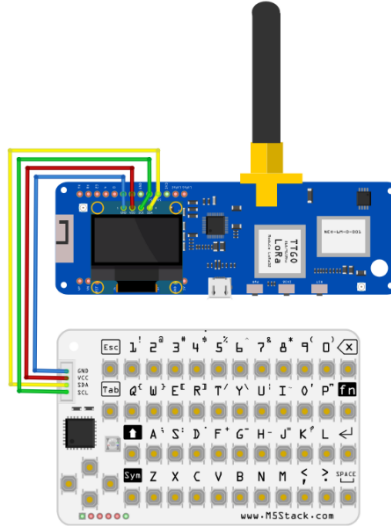


Fig. 5. Node connection diagram

Since the HELTEC Wireless Stick Lite V3 nodes do not have a built-in display and GPS receiver, an external OLED display, GPS receiver and M5STACK CardKB keyboard were added.

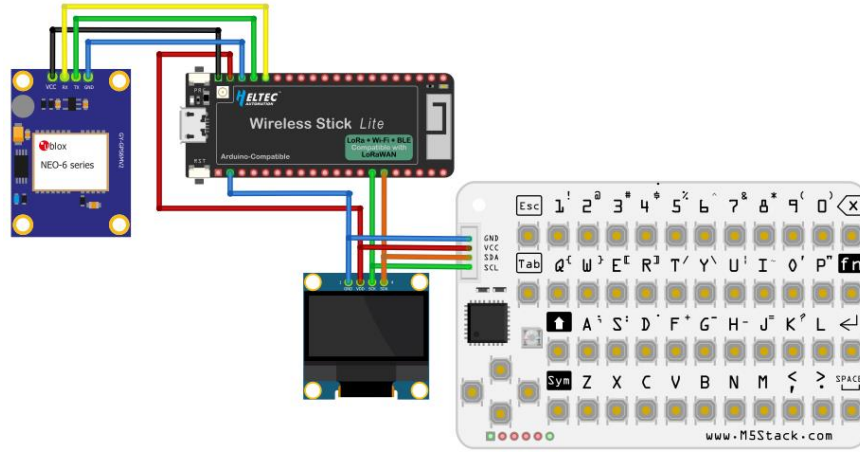


Fig. 6. Node connection diagram

2.1. Radio and Module Configurations

First, to install the latest firmware for the Meshtastic node, visit the website <https://flasher.meshtastic.org/>. Select the brand and model of your node, select stable or unstable firmware, and then install the software through the port connected to your computer.

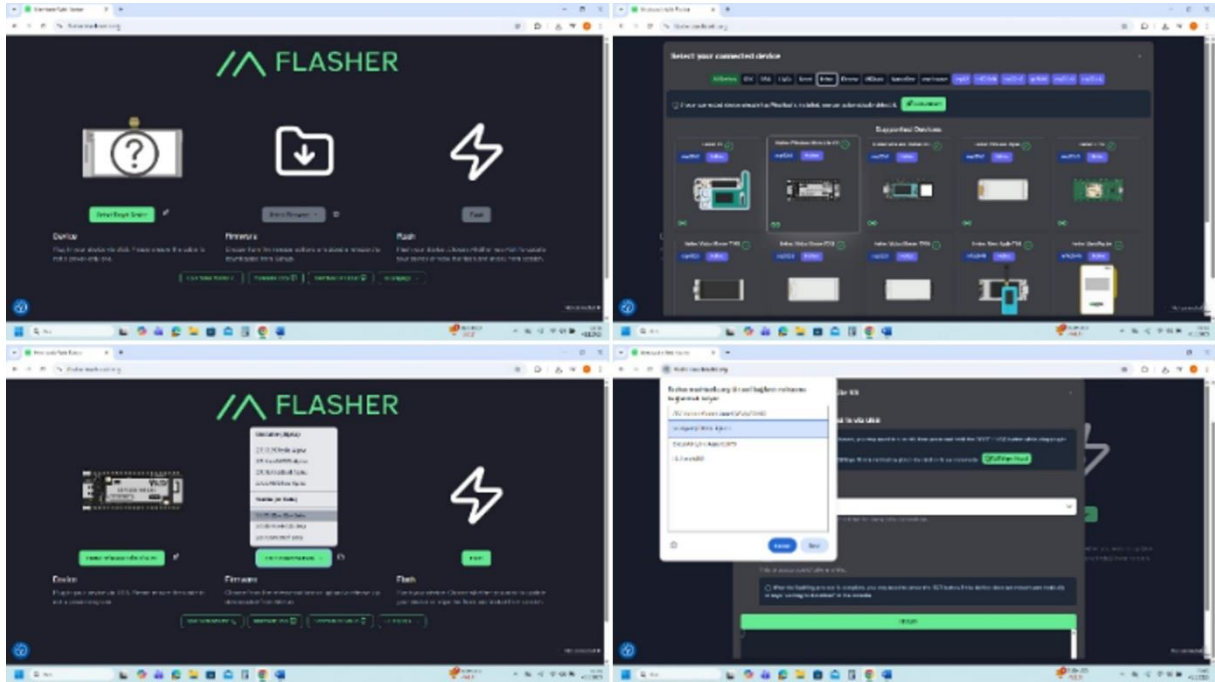


Fig. 7. Meshtastic Flasher installation pages

Using the Android- or IOS-compatible Meshtastic app, the node is connected via Bluetooth and define long and short names are defined in the default Node ID field. Licensed radio amateurs can also activate the corresponding option to lift emission restrictions.

In the LoRa configuration, the nodes are configured with hardware clock speeds of 433MHz and 868MHz, and since they will be used in Europe, the European Union 433/868MHz, Long Range - Fast by default, and the Ok MQTT feature was enabled to receive messages from the Meshtastic MQTT server, and the maximum hop count for messages was set to 3. The default channel settings are 0. The Uplink, Downlink, and Position Enable features were enabled on the channel, enabling the LoRa radio to receive and send messages. The device relay, Client, was selected from the Option property.

Position Packet and Device GPS features were enabled because Meshtastic nodes have a GPS receiver and OLED display. To connect to the MQTT network, an available wireless network is connected using the Wi-Fi option.

The image displays three sequential screenshots of a mobile application for configuring LoRaWAN channels. The first screenshot shows the 'Channels' screen with a list of channels including 'LongFast'. The second screenshot shows the 'Options' for the 'LongFast' channel, including settings for 'PSK', 'Uplink enabled', 'Downlink enabled', 'Position enabled', and 'Precise location'. The third screenshot shows the 'Options' for the 'LongFast' channel, including settings for 'Client', 'Modem', 'Hardware', and 'LED Heartbeat'.

The figure displays four screenshots of the iMoo360 application's settings menu, arranged in a 2x2 grid. Each screenshot shows a different category of settings:

- Position:** Includes 'Broadcast Interval' (set to Five Minutes), 'Smart Position' (toggle on), and 'Fixed Position' (toggle on). It also shows a 'Minimum Interval' of Fifteen Seconds and a 'Minimum Distance' of 100.
- Power:** Features 'Enable power saving mode' (toggle on), 'Flashlight on power loss' (set to Off), 'ADC multiplier override' (toggle on), 'Wait for bluetooth duration' (set to One Minute), 'Super deep sleep duration' (set to Off), and 'Minimum wake time' (set to Ten Seconds).
- Network:** Shows 'WiFi Options' (toggle on), 'Use 12h clock format' (toggle on), 'Bold heading' (toggle on), and 'Display units' (set to METRIC). It also includes a 'Scan WiFi QR code' button and 'UDP Config' (toggle on).
- Display:** Includes 'Always point north' (toggle on), 'Use 12h clock format' (toggle on), 'Bold heading' (toggle on), and 'Display units' (set to METRIC). It also shows 'Advanced' settings like 'Screen on for' (set to Ten Minutes) and 'Carousel interval' (set to Off).

To receive and send messages from the Meshtastic network, the global MQTT server feature was activated and its default features were accepted.

LED, Buzzer, and Vibration can be added to external notification settings to notify when messages arrive via the MQTT server or LoRa. Buttons can be added to send predefined messages, and an external keyboard can be added to type direct messages, as in this example. The input source for the M5stack keyboard is cardkb.

By adding an external detection sensor to a Meshtastic node, messages can be sent upon detection. It is used to send 0-hop neighbor information to the network via neighbor node information. Voice messages can be sent via nodes equipped with Lilygo TLora or LilygoT3S3 LoRa radios. The pxcounter feature is used to determine the number of nodes in the RF network by detecting the MAC addresses of surrounding wireless devices.

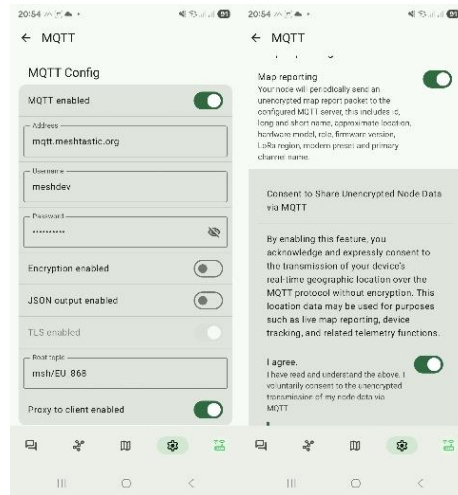


Fig. 11. Meshtastic app page

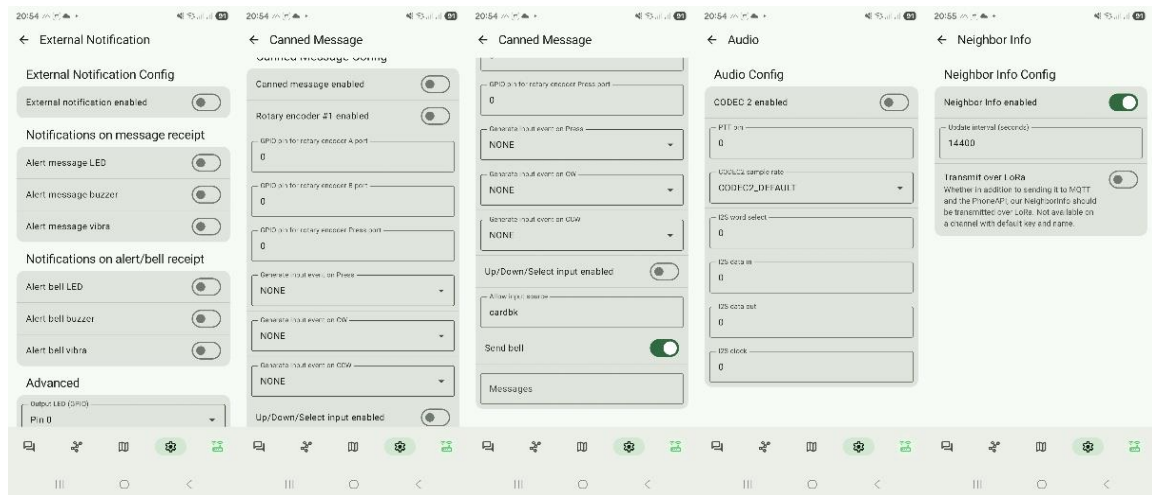


Fig. 12. Meshtastic app page

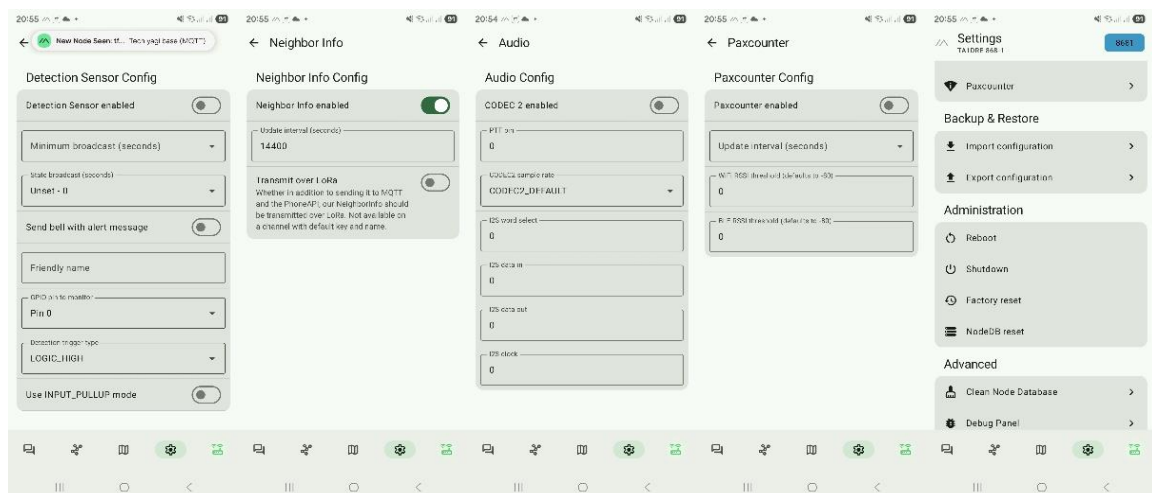


Fig. 13. Meshtastic app page

All settings are saved as backups using the Export Configuration settings and imported into other nodes, preventing the need to re-configure the same settings individually.

Figure 14 shows the nodes prepared for this study, which will operate at frequencies of 433MHz and 868MHz.



Fig. 14. Nodes used in this study

The nodes were given the abbreviations T433, 4331, 4332, T868, 8681, and 8682, respectively. The configuration of all nodes in this study enabled the MQTT server and Map reports to ensure wireless internet access and to display the nodes on the map. Since the internet and MQTT server were enabled, nodes could communicate among themselves via the MQTT server, regardless of radio frequency. Initially, while the internet network was active, messages were sent from each node in the form of “Im T433”, “Im 4331”, “Im 4332”, “Im T868”, “Im 8681”, and “Im 8682”.

The map views of the 433MHz nodes, coordinate data received along the route, and the active nodes in the LongFast channel are shown in Figure 15.

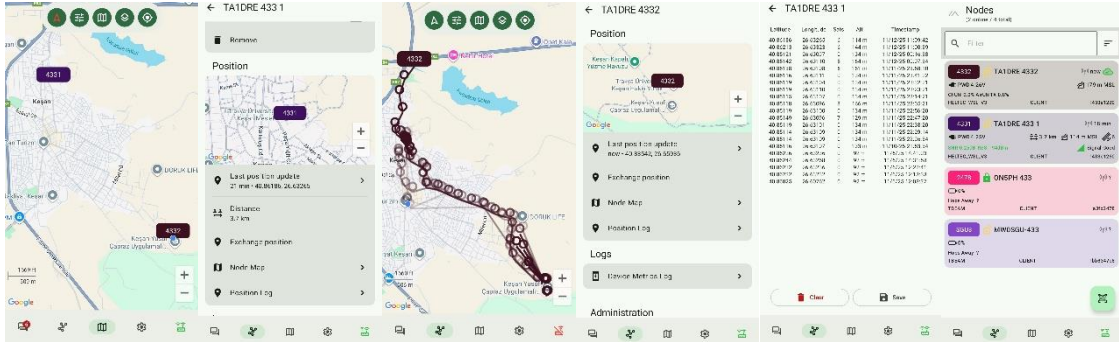


Fig. 15. 433MHz nodes on the map

The map views of the 868MHz nodes and the coordinate data received along the route and the active nodes in the LongFast channel are shown in Figure 16.

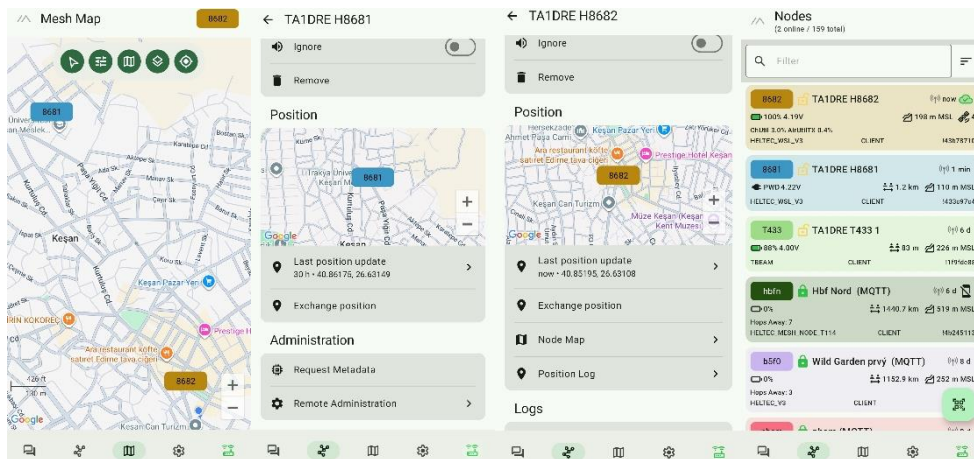


Fig. 16. 868MHz nodes on the map

The internet connection was then disconnected, and the same messages were repeated to test whether the nodes were communicating only via radio frequency. However, since the nodes' MQTT server and map reporting did not work due to the lack of internet connection, screenshots of the nodes were taken and recorded. The messages "Im T433", "Im 4331", "Im 4332", "Im T868", "Im 8681", and "Im 8682" were resent from nodes T433, 4331, 4332, T868, 8681, and 8682, respectively. Since the nodes lacked internet and MQTT connectivity, the sent messages could only be viewed on the screens of other nodes on the same frequency; therefore, screenshots of the node screens were taken. In these tests, when the nodes had a direct and unobstructed view of each other, the Signal-to-Noise Ratio (SNR) was observed to be between 10.0 and 13.0 dB at a distance of 100 meters, and between -10.0 and -11.0 dB at a distance of 1000 meters.

The message sent from node 4331 to nodes T433 and 4332 is shown in the message images displayed on the screens of these nodes in Figure 17.



Fig. 17. Screenshots of this message sent from node 4331 and received from other nodes

The message sent from node 4332 to nodes T433 and 4331 is shown in the message images displayed on the screens of these nodes in Figure 18.



Fig. 18. Screenshots of this message sent from node 4332 and received from other nodes

The message sent from node 8681 to nodes T868 and 8682 is shown in the message images displayed on the screens of these nodes in Figure 19.



Fig. 19. Screenshots of this message sent from node 8681 and received from other nodes

The message sent from node 8682 to nodes T868 and 8681 is shown in the message images displayed on the screens of these nodes in Figure 20.



Fig. 20. Screenshots of this message sent from node 8682 and received from other nodes

3. Conclusion

The Meshtastic application is an open-source project designed to enable text-based communication (voice communication is also possible with some modules) between nodes by hopping between them without using the existing communication infrastructure. In this study, the prepared nodes initially communicated successfully not only among themselves but also with all nodes on the same channel connected to the MQTT server, using internet infrastructure and MQTT server support. Later, the nodes also communicated successfully with each other on their own radio frequencies without an internet connection. Furthermore, by enabling Wi-Fi and disabling the MQTT server feature, communication with other nodes was also established via the node's radio frequency by connecting to the node with smart mobile devices. Since the present study mainly aims to establish communication solely via radio frequency in disaster areas damaged by natural disasters, without using the existing communication infrastructure, especially the internet infrastructure, increasing the number of nodes, using higher-gain antennas, and running nodes in a Repeater role will help increase communication capacity and expand the coverage area of the nodes to ensure uninterrupted communication. Furthermore, since Meshtastic nodes use the Industrial, Scientific, and Medical Devices band, which does not require additional permits or licenses, it will greatly contribute to communication by making it available to anyone who needs it in the event of any disaster. The small and portable nature of the nodes, along with their ability to communicate with computers, tablets, and phones via Wi-Fi, Bluetooth, and USB (serial) port connections in addition to radio frequency, makes Meshtastic an alternative solution to existing communication channels in disaster emergency communication.

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